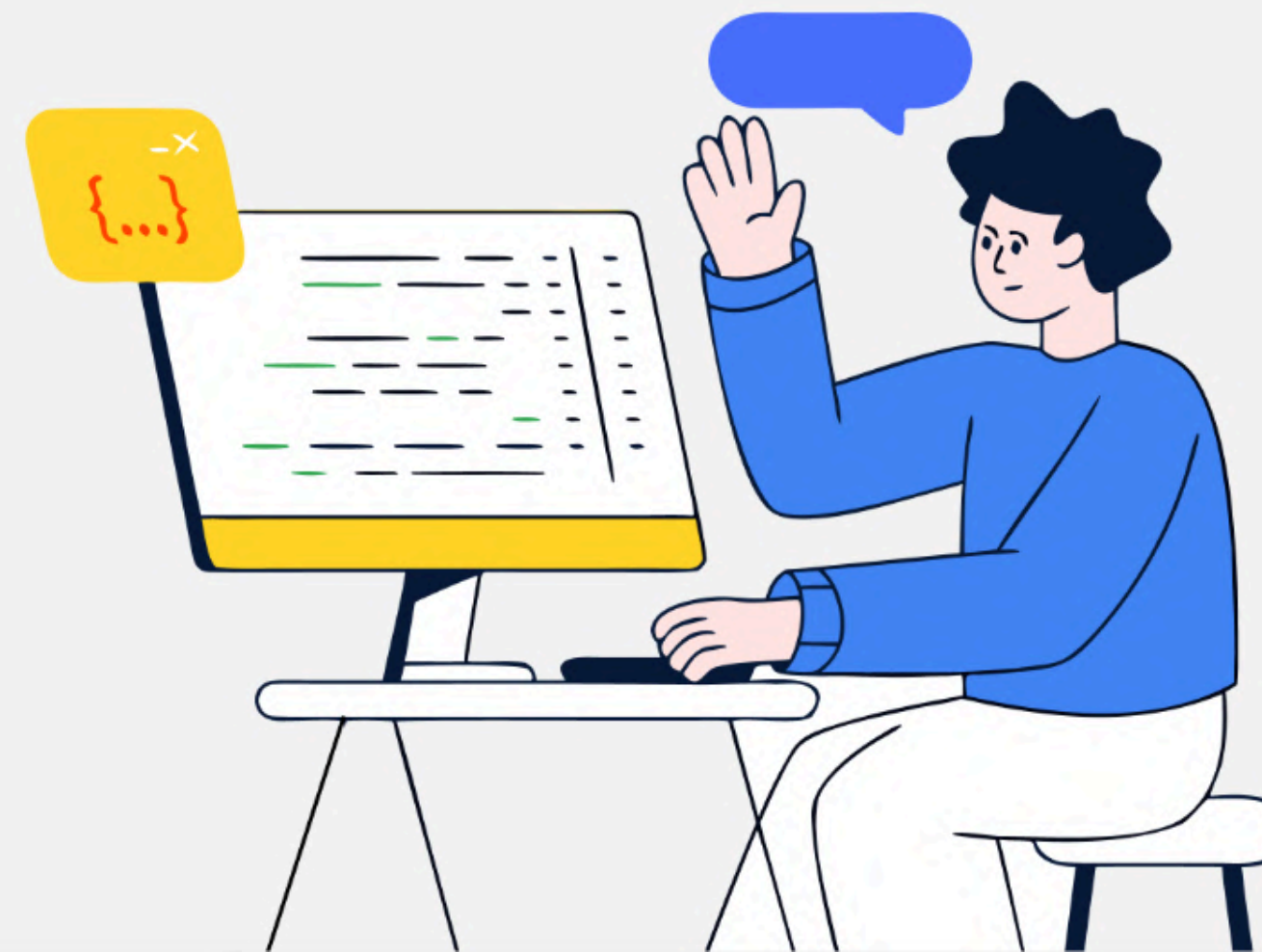




# Solution Challenge



## Team Details

- a. Team name: DEADLOCK
- b. Team leader name: ANISH KUMAR
- c. Problem Statement:  
Rapid Crisis Response  
(Accelerated emergency Response &  
Crisis coordination in hospitality)



EvacuAid

# What Was the Issue?

Today's buildings are designed for *static* emergencies—but crises are inherently *dynamic*. Here is where the current system fails:

## Static Maps, Dynamic Crises

Traditional "fire exit" floor plans don't adapt. When a pre-planned escape route is blocked by smoke, fire, or structural collapse, there is no system to recalculate - leaving people **trapped by the very protocol meant to protect them**.

## Fragmented Situational Awareness

Building administrators and first responders rely on isolated, siloed systems - a single alarm panel here, a manual headcount there. There is no unified, real-time picture of where hazards are and where people are. That information gap costs lives.

## Alarms That Ring, But Don't Think

Current emergency systems are reactive and passive - they alert, but do not orchestrate. No central intelligence processes live sensor data, models crowd flow, or prevents the stampedes that turn evacuations into secondary disasters.

# Brief About Our Solution

- We replace chaos and guesswork with **absolute clarity** by putting an active, adapting escape plan directly in the palm of every user's hand.
- **EvacuAid** is a centralized, AI-orchestrated crisis-management platform that transforms emergency evacuation from a static protocol into a dynamic, real-time intelligence system — saving lives when every second counts.

## Dynamic Evacuation Engine

A custom A\* Pathfinding Algorithm instantly recalculates hazard-free evacuation routes the moment an IoT sensor is triggered — no static maps, no guesswork.

## Cross-Platform Real-Time Sync

A glassmorphic **Next.js Web Command Centre** for admins, perfectly synchronised with a **React Native Mobile App** for on-ground first responders — on a single unified API.

## Gemini AI Orchestration

Integrated Google Gemini Pro acts as a Safety Assistant Node — parsing raw sensor data and auto-broadcasting human-readable crisis protocols in real time.

## Enterprise-Grade Resilience

Deployed on Vercel and anchored by Google Cloud SQL — delivering 99.99% uptime and unyielding data persistence precisely when seconds matter most.

# EvacuAid: Not Just Smarter — Fundamentally Different

Legacy static systems were built for a different era. EvacuAid literally redraws the map in real time.

## The Traditional Approach

### Static Fire Maps

Printed before the fire even starts. Useless once a corridor is blocked.

### Loud Alarms

Trigger panic, provide zero guidance, and leave responders guessing.

### Walkie-Talkie Chatter

Disjointed, delayed, prone to miscommunication under extreme stress.

## The EvacuAid Way

### A\* Topological Pathfinding

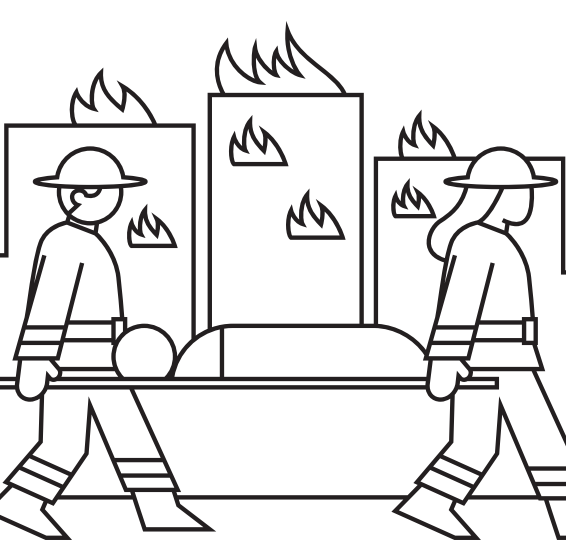
Mathematically recomputes the safest escape vector as the fire spreads — in real time.

### Google Gemini AI

Reads raw sensor alarms and instantly translates them into plain-text evacuation protocols.

### Proactive & Dynamic

No other system actively redraws physical evacuation vectors based on live hazard propagation.



### ❑ The Core USP:

EvacuAid is the only solution that combines A\* topological pathfinding with Google Gemini AI to dynamically reroute evacuees around a *moving* hazard — not a static one.



# How EvacuAid Solves the Problem?

## Eliminating Guesswork with A\*

The **A\* algorithm** conceptualizes the building as a **weighted graph**.

Each **node represents a room or corridor**, while each edge bears a dynamic weight informed by sensor data. As fire propagates, edges are either penalized or eliminated, prompting the algorithm to rapidly recalculate the most efficient escape route. In this way, human panic gives way to mathematical precision.

### A\* Pathfinding Core

Topological algorithm computing optimal live escape vectors

## One Source of Truth via Google Cloud SQL

The moment a hazard ping lands in our Google Cloud SQL database, both the Web Dashboard (for incident commanders) and the Mobile App (for ground responders) update simultaneously. Every stakeholder – regardless of device or location – sees the same live picture. The communication gap is permanently closed.

2

### Platforms Synced

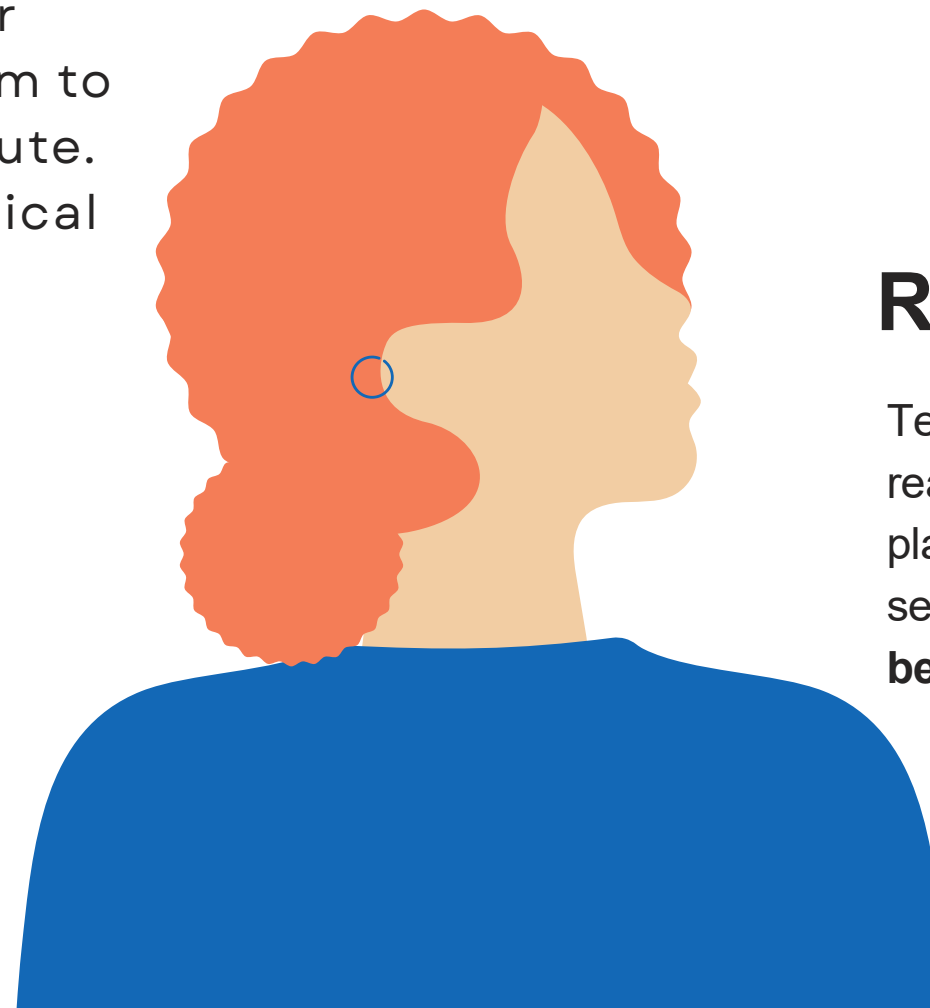
Web Dashboard + Mobile App  
from a single Cloud SQL source

## Raw Data(Human Strategy via Gemini)

Ten flashing sensor lights mean nothing under stress. **Google Gemini** reads every active alarm, interprets the spatial context, and broadcasts a plain-language evacuation protocol to all connected devices. Complex sensor matrices become clear, actionable commands in seconds. **Data becomes decisions.**

### 0 Guesswork

Every decision is mathematically computed and AI-narrated in real time



# Features of EvacuAID

Six purpose built capabilities that unify real-time hardware intelligence, AI decision-making, and cross-platform coordination.

## ***Dynamic Pathfinding Engine***

Uses graph topology to instantly redraw evacuation routes around active hazard zones in real time, ensuring responders always have a safe, updated path.

## ***Google Gemini Responder AI***

An integrated floating chat node that ingests raw building data and translates it into clear, human-readable evacuation strategies on demand.

## ***Omnibox Command Palette***

A global Cmd+K dashboard that lets admins deep-search IoT sensors, active incidents, and locations across the entire system in milliseconds.

## ***Cross-Platform Telemetry***

Instant, automatic synchronisation between the Next.js Admin Dashboard and the React Native Responder Mobile App – zero data lag under pressure.

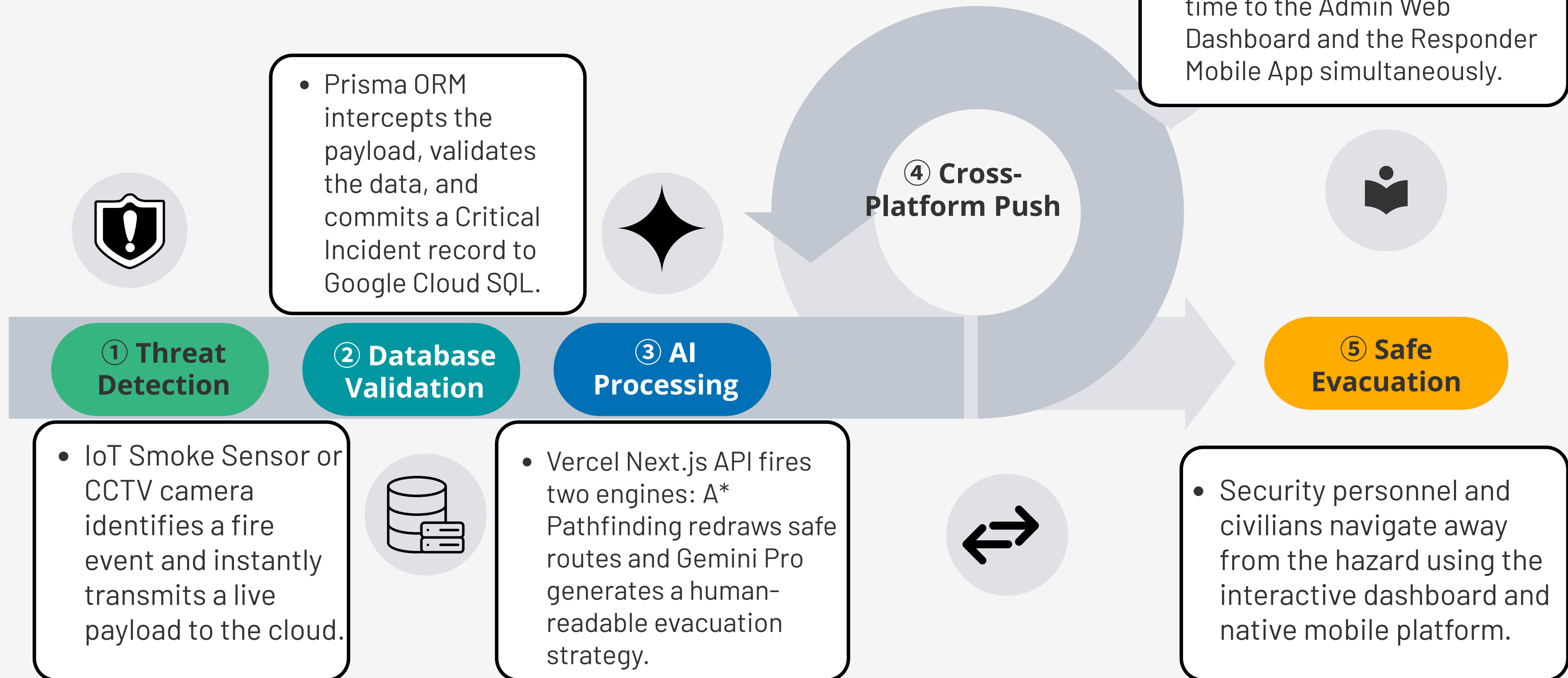
## ***IoT Hardware Integration***

Live connection and real-time tracking of physical devices including CCTV cameras and smoke/fire detectors directly within the command interface.

## ***Prisma & Google Cloud Architecture***

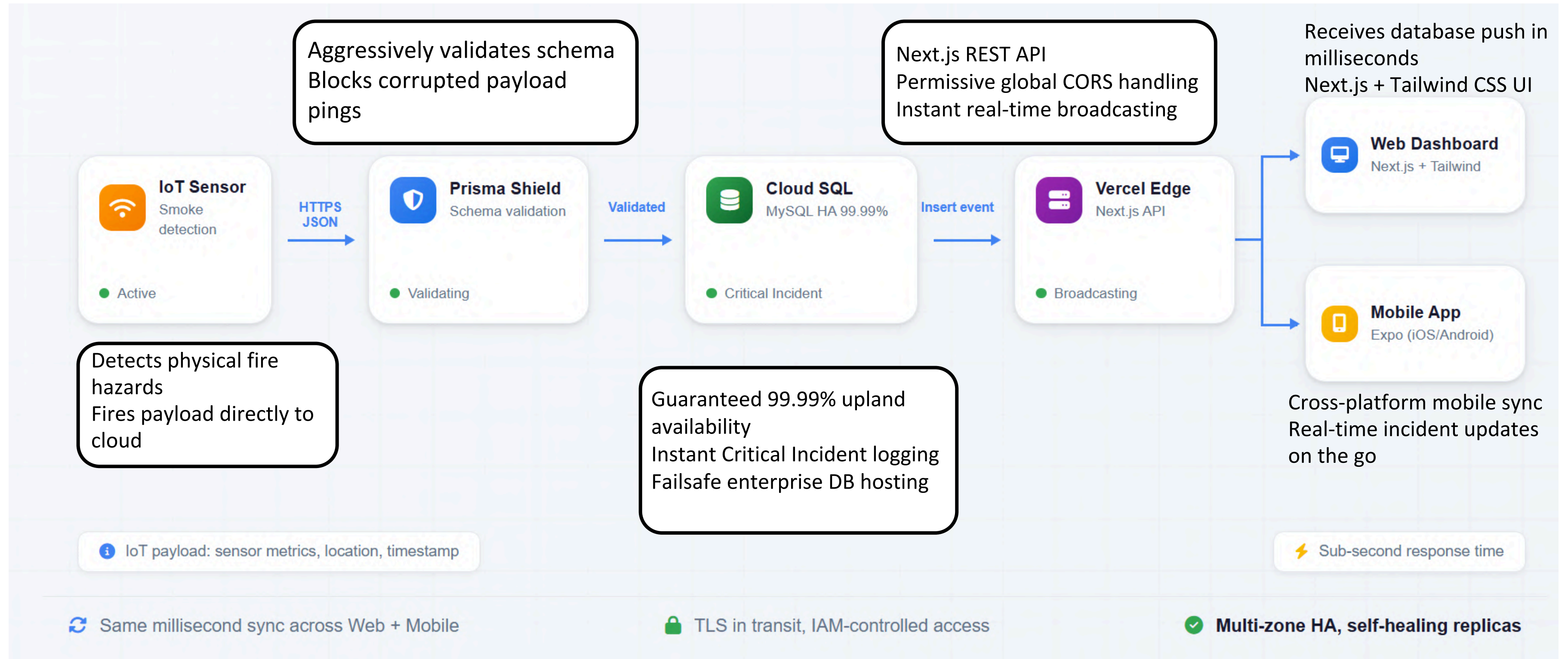
A resilient backend built on Prisma ORM validation backed by Google Cloud SQL, engineered for 99.99% uptime during critical emergencies.

# PROCESS FLOW DIAGRAM



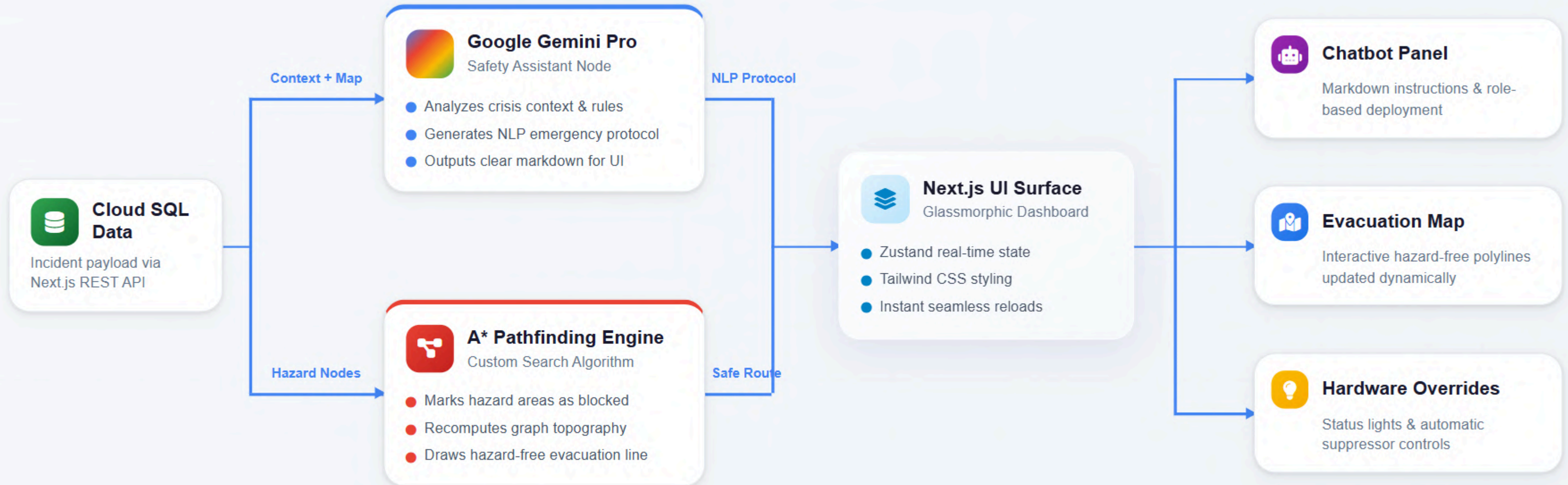


# Architecture diagram of the proposed solution

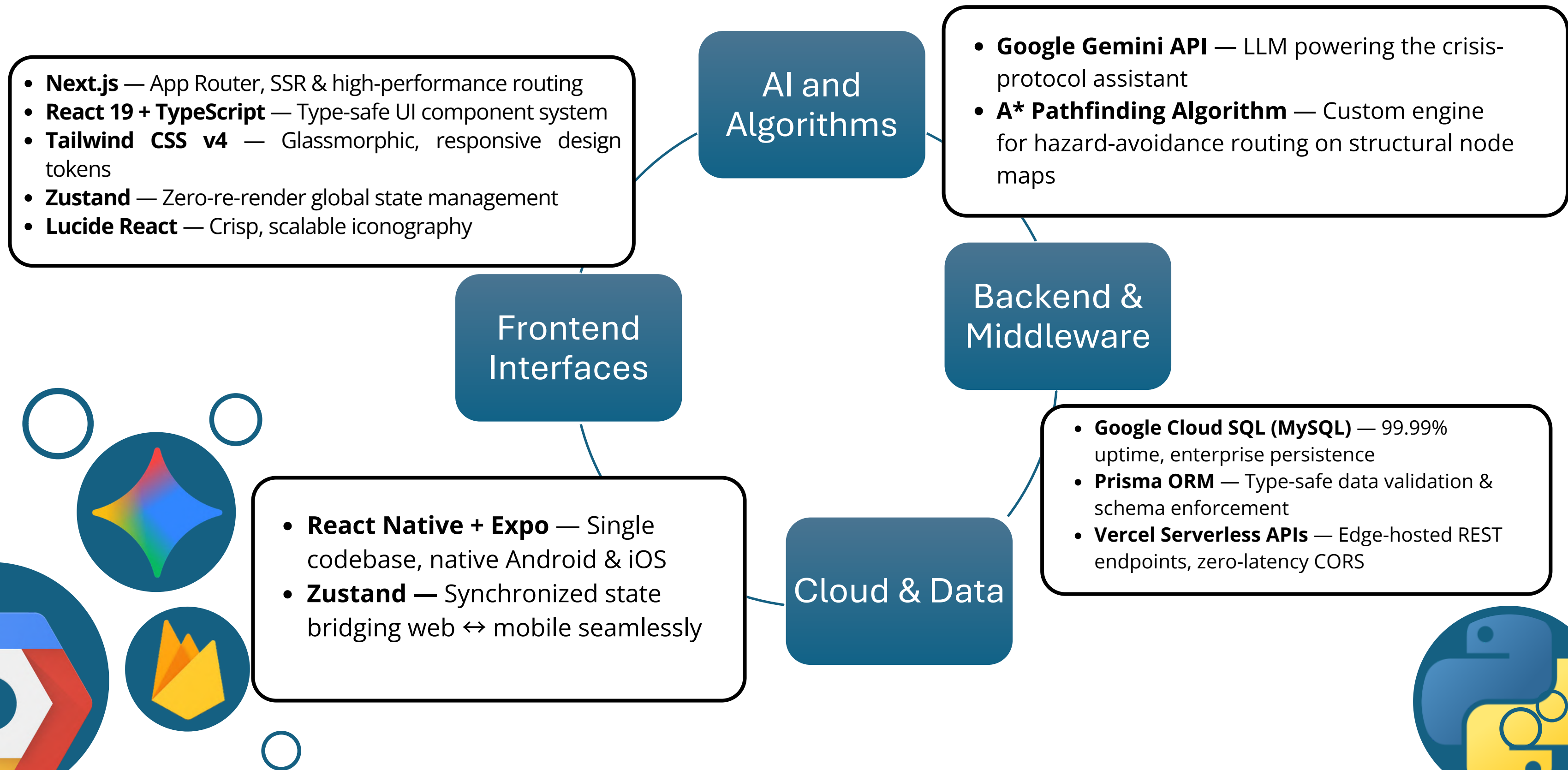




## Architecture diagram of the proposed solution (continued...)



## • Technologies & Architecture Stack



# Estimated Implementation Cost

A lean, serverless architecture means EvacuAid scales from a single pilot to a full commercial deployment — without the burden of idle infrastructure costs.

## LEVEL 1

### Prototype & Pilot

**Vercel Edge Network:** \$0/mo — Hobby Plan

**Google Gemini API:** \$0/mo — Free tier

**Google Cloud SQL:** ~\$15/mo — db-f1-micro

**React Native App:** \$0 — Open-source sideload

**ESP32 / Pi Gateway:** \$15–\$35 one-time

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**Total:** ~\$15/month + marginal hardware

## LEVEL 2

### Commercial Scale Deployment

**Vercel Pro:** \$20/mo — High-traffic bandwidth

**Gemini Pro API:** \$10–\$25/mo — Pay-as-you-go tokens

**Cloud SQL (Dedicated vCPU):** \$50–\$80/mo

**App Store (Apple Dev):** \$99/year

**IoT Retrofitting:** \$0 if IP-native; else ~\$20/module

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**Total:** ~\$100–\$150/month software + variable hardware

✅ **The Serverless Advantage:** EvacuAid pays only for compute consumed during an active emergency — not 24/7 idle server uptime. This makes it exponentially more affordable than legacy enterprise hardware systems that require constant maintenance, dedicated server rooms, and six-figure deployment contracts.



# Snapshots of the MVP

Search incidents or commands...

EN  System

## Command Dashboard

Real-time overview of all active operations. [3 emergencies on Building Map](#)

**ACTIVE INCIDENTS**  
**7**  
3 from Building Map

**CRITICAL ALERTS**  
**4**  
Requires immediate attention

**PENDING TASKS**  
**5**  
1 resolved today

**UNREAD ALERTS**  
**3**  
50% ACK rate overall

### Recent Incidents

Live anomaly log [All Incidents →](#)

**INC-2053**  **CRITICAL**  
**Fire**  
First Floor – South Corridor 11:24 PM

**INC-2052**  **CRITICAL**  
**Fire**  
Ground Floor – Room 110 11:24 PM

**INC-2051**  **CRITICAL**  
**Fire**  
Ground Floor – Room 111 06:28 AM

**INC-2045** **LOW**  
**IT Offline**  
Sector 2, Fl 1 11:30 AM

**INC-2044** **LOW**  
**Maintenance**  
Basement HVAC 11:15 AM

### PRIORITY ALERTS

11:52:46 PM

**CRITICAL** 11:24 PM  
**Fire — First Floor – South Corridor**  
Fire / smoke detected by building map sensor.  
Auto-Escalation in: **01:35**

**CRITICAL** 11:24 PM  
**Fire — Ground Floor – Room 110**  
Fire / smoke detected by building map sensor.  
[View All 7 Active Incidents →](#)

### Pending Tasks

5 open assignments [Task Board →](#)

**TSK-103** → **INC-2053**  
Respond to Fire at First Floor – South Corridor **High**

**TSK-101** → **INC-2051**  
Respond to Fire at Ground Floor – Room 111 **High**

**TSK-094** → **INC-2044**  
Investigate HVAC fault in basement **Low**

### Notification Center

3 unread · 50% ACK rate

**Fire Dispatch** **FIRE DEPT**  
**INC-2053**  
[INC-2053] Fire at First Floor – South Corridor — Fire /

**Fire Dispatch** **FIRE DEPT**  
**INC-2051**  
[INC-2051] Fire at Ground Floor – Room 111 — Fire / sm

**Dr. L. Evans** **MEDICAL STAFF**  
**INC-2042**  
[INC-2042] Medical emergency in Lobby Area - awaiti

 **Building Map**  
A\* Facility Routing Engine

## EvacuAid

- Dashboard
- Building Map
- Incidents
- Tasks
- Notifications
- Analytics
- IoT Devices**
- Settings

## IoT Command Center

Monitor and control physical facility hardware sensors.

**TOTAL MANAGED**  
**12**

**ACTIVE STREAMS**  
**7**

**OFFLINE**  
**2**

[All Hardware](#) [CCTV Feeds](#) [Sensors / Doors](#)

### Live Visual Feeds

**Lobby Cam 01** **ONLINE**  
**REC**

**Corridor Cam North** **ONLINE**  
**REC**

**Server Room Cam**  
**REC**

## EvacuAid

Search incidents or commands...

EN  System Online

[Exit Map to Dashboard](#)

[Map Editor](#) [Navigate & Route](#) **4 Critical** **7 Active** [Step-by-Step Directions](#)

**Routing Controls**

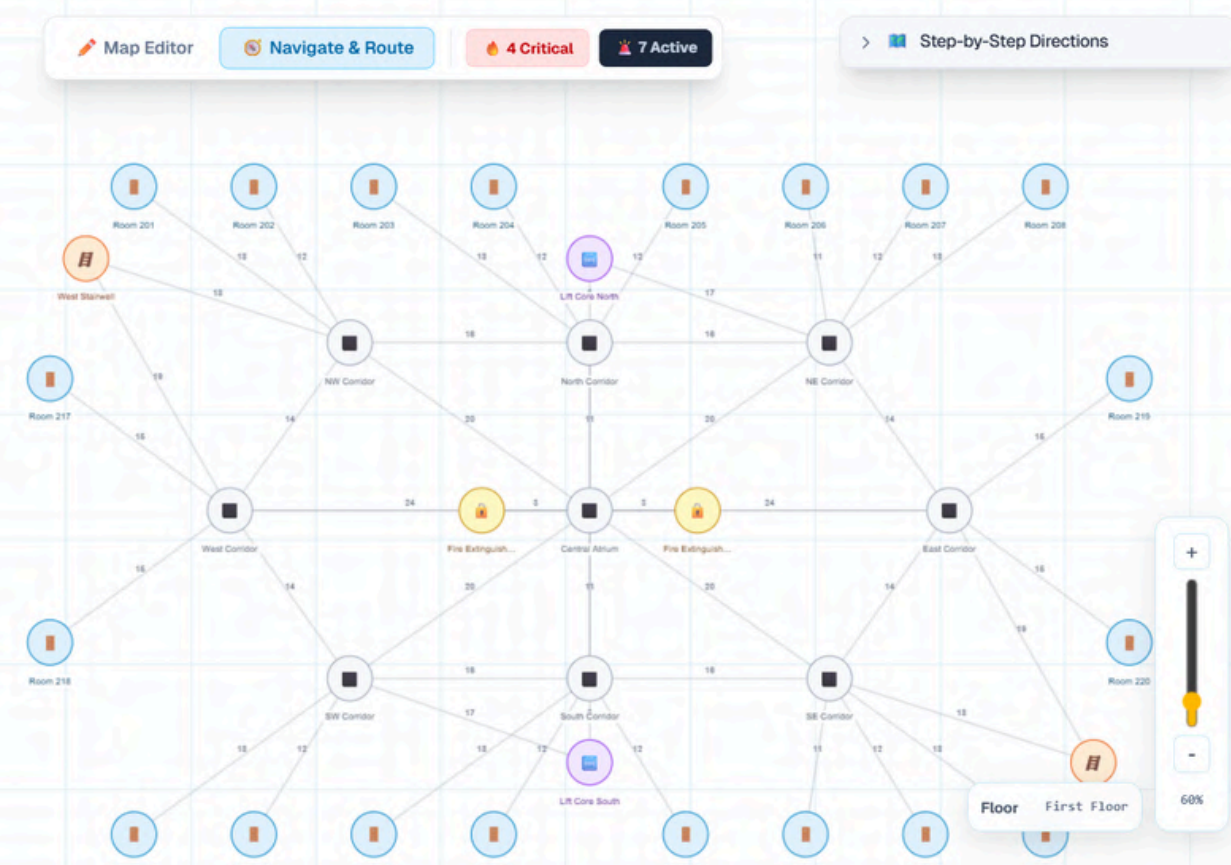
**SCENARIO**

- Complex Normal Ops**  
4 floors, 20 rooms per floor, dual stairwells, dual lifts, and extinguisher points on every floor.
- Fire Emergency (Floor 3)**  
North-east cluster on Floor 3 is hazardous and partially blocked to force meaningful rerouting.
- Medical Response (Floor 2 South Wing)**  
South corridor on Floor 2 has elevated risk cost to test danger-aware path selection.
- Lift Maintenance Bottleneck**  
South lift core is unavailable, forcing stair/lift-north alternatives across the building.

**START LOCATION**  
Room 207

**DESTINATION**  
Emergency Exit

**ALGORITHM**



Floor First Floor

## Additional Details & Future Development

EvacuAid is fully operational today — but this roadmap represents our mission to set the industry standard for intelligent, life-saving building safety.

### AR Indoor Navigation

Google ARCore projects A\*-calculated arrows onto physical floors — guiding responders through smoke-filled corridors in real time.

### Edge-AI Computer Vision

Lightweight ML models run directly on CCTV hardware — detecting intruders, collapses, or spreading fires in milliseconds, triggering our Next.js API instantly.

### City-Wide Dispatch Link

Vercel Serverless APIs handshake with municipal 911 centres — automatically streaming live building maps and Gemini protocols to approaching fire engine tablets.

### Wearable Haptic Feedback

REST API pushes reach Wear OS smartwatches — physically buzzing civilians and pointing them toward the nearest unblocked stairwell, even without audio.

### Predictive Crowd Simulation

Historic Google Cloud metrics train our A\* algorithm to anticipate choke points — preemptively splitting evacuation crowds across multiple routes to prevent stampedes.



Links to our solution prototype:

1. GitHub Public Repository

i. WEBSITE: <https://github.com/Anish-ai/EvacuAid>

ii. APPLICATION: <https://github.com/Anish-ai/EvacuAid-app>

2. Demo Video Link(Detailed Demo):

i. <https://www.youtube.com/watch?v=9pvvVtZFjOc>

3. Demo Video Link(Short Demo):

i. <https://www.youtube.com/watch?v=zIKpQWPV4GM>

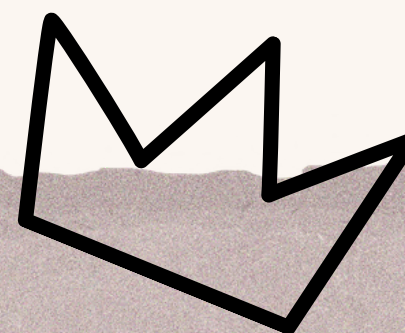
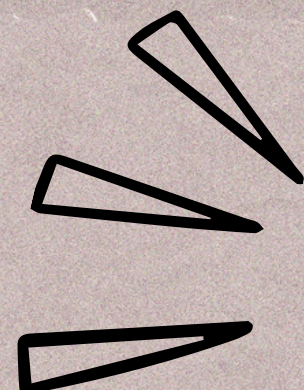
4. MVP Link

i. WEBSITE: <https://evacu-aid.vercel.app>

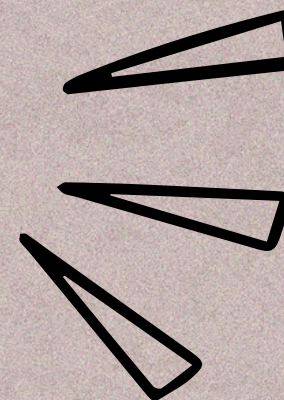
ii. Application: [https://drive.google.com/drive/folders/16v-aqoCCzmj8oASdVR\\_E69--3zk6U9JU](https://drive.google.com/drive/folders/16v-aqoCCzmj8oASdVR_E69--3zk6U9JU)

5. Working Prototype Link: <https://evacu-aid.vercel.app>





# Thank you



**Team DEADLOCK**

- ANISH KUMAR
- Kushal Kesharwani
- Jatin Kumar
- Ankit Kumar